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HT Board Booster

Class X Science

Physics

CBSE Board Paper Question Bank

Chapter Wise Questions

Ch - 2 Human Eye

Session 2026-27



Class X Science Physics
Ch- Human Eye
Previous Year CBSE Board Question Paper

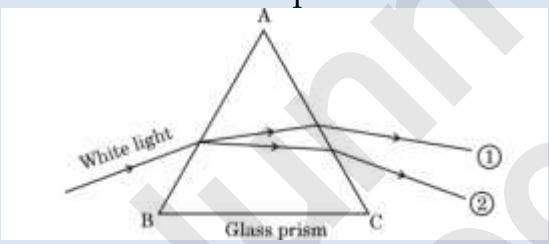
CBSE BP 2025-26
Section-A

Q.1. When you look at an object very close to your eyes, the: (Set- 31/1/1)
(a) Ciliary muscles of your eye contract and the eye lens become thick.
(b) Ciliary muscles of your eye get relaxed and the eye lens becomes thick.
(c) Ciliary muscles of your eye contract and the eye lens becomes thin.
(d) Ciliary muscles of your eye get relaxed and the eye lens becomes thin.

Q.2. **Assertion (A):** When rays of white light pass through a prism, on emerging they give spectrum of seven colours. (Set- 31/1/1)
Reason (R): It is due to the scattering of light that red light bends minimum and violet light bends the maximum.

Q.3. A thin beam of white light falls on a face of a glass prism and a spectrum is obtained on a screen. Which of the following options is the correct sequence of colours from top to bottom on the screen? (Set- 31/2/1)
(a) Red, Green, Yellow, Violet, Blue
(b) Red, Yellow, Green, Blue, Violet
(c) Violet, Blue, Green, Yellow, Red
(d) Violet, Red, Green, Yellow, Blue

Q.4. Observe the given ray diagram showing refraction through a prism and choose the correct option for colour of rays 1 and 2 : (Set- 31/3/1)



Colour of ray 1	**Colour of ray 2**
(A) Blue	Red
(B)Green	Yellow
(C)Red	Violet
(D)Violet	Red

Q.5. The pair of eye parts responsible for admitting different amount of light in to the eyes is
(a) Iris and pupil (b) Ciliary muscles and pupil (Set- 31/4/1)
(c) Retina and Iris (d) Ciliary muscles and cornea

Q.6. The vision defect which arises due to gradual weakening of the ciliary muscles and diminishing flexibility of eye lens is: (Set- 31/4/1)
(a) Myopia (b) Hypermetropia
(c) Presbyopia (d) Myopia and hypermetropia both

Q.7. To restore clear vision in persons whose size of the eye ball has reduced, he/she is suggested to use suitable (Set- 31/5/1)
(a) Converging lens (b) Diverging lens
(c) Bifocal lens (d) Cylindrical lens

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Section-B

Q.8. Attempt either (a) or (b) : (Set- 31/3/1)

(a) What happens when light is scattered from (i) very fine particles and (ii) very large size particles of medium ? Justify your answer in each case.

OR

(b) With the help of a ray diagram, explain any phenomenon of formation of spectrum in nature.

Q.9. (i) How does the change in curvature of the eye lens helps us in the process of seeing the nearby objects clearly? (Set- 31/5/1)

(ii) State the range of the power of accommodation of a normal human eye.

OR

Draw a ray diagram to show the correction of eye defect of an old man who can not see an object placed closer than 1 m from his eye, clearly.

Q.10. (a) Draw the ray diagram for refraction of light through a glass prism and mark angle of refraction and angle of deviation. (Set- 31/5/1)

(b) When the path of a light ray refracted through a glass prism is reversed how will the angle of deviation change? Explain.

Section-C

Q.11. A person is unable to read a book placed closer than 1 meter from his eyes. Identify the defect of vision in his eyes. Draw the ray diagrams to show the defect of vision and its correction. (Set-31/1/1)

Q.12. Attempt either (a) or (b): (Set- 31/2/1)

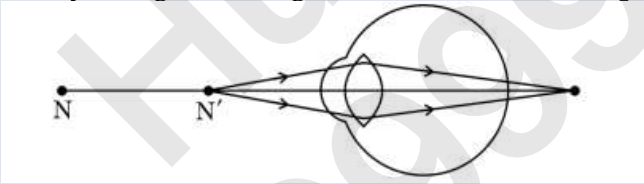
(a) How does the focal length of eye lens change as the distance of the object from the eyes is altered? Explain.

OR

(b) Why is the concave lens used as a corrective lens for a myopic eye?

Q.13. What is scattering of light? How is scattering of light different from reflection of light? Explain. (Set- 31/2/1)

Q.14. Study the given diagram and answer the questions that follow : (Set- 31/3/1)



(a) Write the name of the eye defect shown in the diagram. Where is the image formed in this eye defect with respect to the retina of the eye?

(b) List two causes of this eye defect.

(c) With the help of a diagram, show how this eye defect of vision is corrected.

Q.15. Explain the phenomenon of rainbow formation in the sky. Draw the labelled ray diagram in support of your explanation. (Set- 31/4/1)

Q.16. (a) Draw the ray diagram for refraction of light through a glass prism and mark angle of refraction and angle of deviation. (Set- 31/5/1)

(b) When the path of a light ray refracted through a glass prism is reversed how will the angle of deviation change? Explain.

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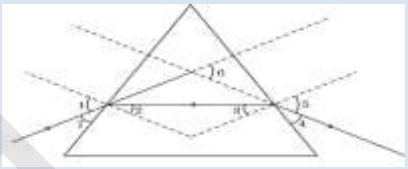
CBSE BP 2024-25

Section-A

Q.17. An old person is suffering from an eye defect caused by weakening of ciliary muscles and diminishing flexibility of the eye lens. If the defect of vision is 'a' which can be corrected by lens 'b', then 'a' and 'b' respectively are: (Set- 31/1/1)

(a) hypermetropia and convex lens (b) presbyopia and bifocal lens
(c) myopia and concave lens (d) myopia and bifocal lens

Q.18. In the given figure the angle of incidence and the angle of deviation respectively are: (Set- 31/2/1/)



(a) 1 and 5 (b) 7 and 6
(c) 7 and 4 (d) 1 and 6

Q.19. When a beam of white light passes through a region of very fine dust particles, the colour of light that scatters the most in that region is: (Set- 31/3/1)

(a) red (b) orange (c) blue (d) yellow

Q.20. **Assertion (A):** When ciliary muscles contract, eye lens becomes thin. (Set- 31/4/1)
Reason (R): Ciliary muscles control the power of the eye lens.

Q.21. The possible way to restore clear vision of those people whose eyeball has elongated is the use of suitable (Set- 31/5/1)

(a) bifocal lens (b) concave lens
(c) converging lens (d) convex lens

Q.22. The part of human eye which controls the amount of light entering into it. (Set- 31/6/1)

(a) Iris (b) Cornea (c) Ciliary muscles (d) Pupil

Section-B

Q.23. (a) A student has difficulty in reading his textbooks but can read the blackboard clearly while sitting in the last row. Name the defect of vision the student is suffering from. List two reasons due to which this defect arises. Write the nature of the lenses required to correct this defect. (Set- 31/3/1)

OR

(b) Draw a ray diagram to show the path of a ray of light falling obliquely on one of the refracting faces of a triangular glass prism and mark the angle of deviation on it.

Q.24. (a) Give reasons: (Set- 31/6/1)
(i) The sky appears dark to passengers flying at very high altitude.
(ii) 'Danger' signal lights are red in colour.

OR

(b) What is a rainbow? "We see a rainbow in the sky only after the rainfall." Why?

Section-C

Q.25. What is a rainbow? Draw a labelled diagram to show its formation. (Set- 31/3/1)

Q.26. A person uses lenses of +2.0 D power in his spectacles for the correction of his vision.
(a) Name the defect of vision the person is suffering from. (Set- 31/6/1)
(b) List two causes of this defect.
(c) Determine the focal length of the lenses used in the spectacles.

Section-D

Q.27. (i) The power of a lens 'X' is 2.5 D. Name the lens and determine its focal length in cm. For which eye defect of vision will an optician prescribe this type of lens as a corrective lens? (Set- 31/1/1)

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(ii) "The value of magnification 'm' for a lens is 2." Using new Cartesian Sign Convention and considering that an object is placed at a distance of 20 cm from the optical centre of this lens, state:

(I) the nature of the image formed;

(II) size of the image compared to the size of the object;

(III) position of the image, and (IV) sign of the height of the image.

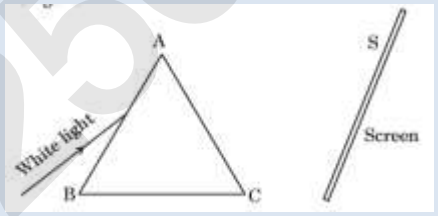
(iii) The numerical values of the focal lengths of two lenses A and B are 10 cm and 20 cm respectively. Which one of the two will show higher degree of convergence/divergence? Give reason to justify your answer.

OR

(i) Draw a ray diagram to show the refraction of a ray of light through a rectangular glass slab when it falls obliquely from air into glass.

Section-E

Q.28. A person allowed a narrow beam of white light from the sun to enter a dark room through a small aperture and placed a glass prism in its path in such a manner that the beam falls on the face AB of the prism as shown in the figure. (Set- 31/2/1)



A screen S is placed on the other side of the prism, facing AC. On turning the prism slowly, a beautiful band of colours is obtained on the screen. It is the spectrum of sunlight.

(a) Name the phenomenon due to which a prism splits the incident white light into a band of colours.

(b) State the reason of getting a band of seven colours in the above case.

(c) (i) Explain with the help of a labelled ray diagram, an experimental arrangement to show the recombination of the spectrum of white light.

OR

(c) (ii) Draw a labelled ray diagram to show the formation of a rainbow.

Q.29. The students in a class took a thick sheet of cardboard and made a small hole in its centre. Sunlight was allowed to fall on this small hole and they obtained a narrow beam of white light. A glass prism was taken and this white light was allowed to fall on one of its faces. The prism was turned slowly until the light that comes out of the opposite face of the prism appeared on the nearby screen. They studied this beautiful band of light and concluded that it is a spectrum of white light. (Set- 31/4/1)

(i) Give any one more instance in which this type of spectrum is observed.

(ii) What happens to white light in the above case?

(iii) (A) List two conditions necessary to observe a rainbow.

OR

(iii) (B) Draw a ray diagram to show the formation of a rainbow. Mark on it, points (a), (b) and (c) as given below:

(a) Where dispersion of light occurs.

(b) Where light gets reflected internally.

(c) Where final refraction occurs.

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Section-A

Q.30. The lens system of human eye forms an image on a light sensitive screen. which is called as: (Set- 31/1/1)
(a) Cornea (b) Ciliary muscles
(c) Optic nerves (d) Retina

Q.31. **Assertion (A):** The rainbow is a natural spectrum of sunlight in the sky. (Set- 31/1/1)
Reason (R): Rainbow is formed in the sky when the sun is overhead and water droplets are also present in air.

Q.32. Consider the following statements in the context of human eye: (Set- 31/2/1)
(a) The diameter of the eye ball is about 2.3 cm.
(b) Iris is a dark muscular diaphragm that controls the size of the pupil.
(c) Most of the refraction for the light rays entering the eye occurs at the crystalline lens.
(d) While focusing on the objects at different distances the distance between the crystalline lens and the retina is adjusted by ciliarymuscles.

Q.33. **Assertion (A)** : Myopic eye cannot see distant objects distinctly. (Set- 31/2/1)
Reason (R) : For the correction of myopia converging lenses of appropriate power are prescribed by eye-surgeons.

Q.34. The Phenomena of light involved in the formation of a rainbow in the sky Are:
(a) Refraction, dispersion and reflection (Set- 31/3/1)
(b) Refraction, dispersion and total internal reflection
(c) Dispersion, scattering and reflection
(d) Dispersion, refraction and internal reflection

Q.35. **Assertion (A):** Red light signals are used to stop the vehicles on the road. (Set- 31/3/1)
Reason (R) : Red coloured light is scattered the most so as to be visible from a large distance.

Q.36. When a beam of white light passes through a region having very fine dust particles, the colour of light mainly scattered in that region is : (Set- 31/4/1)
(a) Red (b) Orange (c) Blue (d) Yellow

Q.37. The phenomena of light involved in the formation of rainbow are : (Set- 31/5/1)
(a) Refraction, reflection and dispersion
(b) Refraction, dispersion and internal reflection
(c) Reflection, dispersion and internal reflection
(d) Refraction, dispersion, scattering and total internal reflection

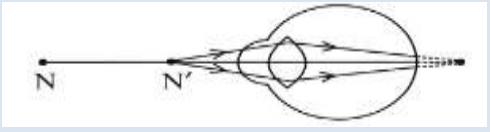
Q.38. The colour of light for which the refractive index of glass is minimum, is : (Set- 31/5/1)
(a) Red (b) Yellow (c) Green (d) Violet

SECTION-B

Q.39. When do we say that a particular person is suffering from hypermetropia? List two causes of this defect. Name the type of lens used to correct this defect. (Set- 31/3/1)

SECTION-C

Q.40. Study the diagram given below and answer the questions that follow: (Set- 31/1/1)



(i) Name the defect of vision represented in the diagram, Give reason for your answer.
(ii) List two causes of this defect.
(iii) With the help of a diagram show how this defect of vision is corrected.

Q.41. (a) Define the term power of accommodation of human eye. Write the name of the part of eye which plays a major role in the process of accommodation and explain what happens when human eye focuses (Set- 31/2/1)

(i) nearby objects and (ii) distant objects.

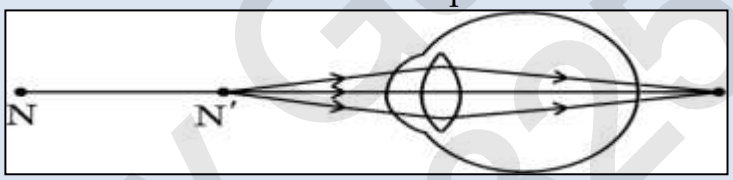
OR

(b) Draw a ray diagram to show the formation of a rainbow in the sky.
On this diagram mark A - where dispersion of light occurs, B - where internal reflection of light occurs and C - where refraction of light occurs. List two necessary conditions to observe a rainbow.

Q.42. Name and explain the phenomenon of light due to which the path of a beam of light becomes visible when it enters a smoke-filled room through a small hole. Also state the dependence of colour of the light we receive on the size of the particle of the medium through which the beam of light passes. (Set- 31/3/1)

Q.43. Define the term power of accommodation of human eye. What happens to the image distance in the eye when we increase the distance of an object from the eye? Name and explain the role of the part of human eye responsible for it in this case. (Set- 31/4/1)

Q.44. Study the diagram given below and answer the questions that follow : (Set- 31/5/1)



(i) Name the defect of vision depicted in this diagram stating the part of the eye responsible for this condition.

(ii) List two causes of this defect.

(iii) Name the type of lens used to correct this defect and state its role in this case.

OR

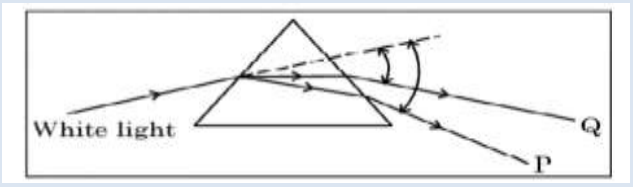
(b) What is dispersion of white light ? State its cause. Draw a diagram to show dispersion of a beam of white light by a glass prism.

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Section-A

Q.45. **Assertion (A):** A person suffering from myopia cannot see the distant objects clearly.
Reason (R): A converging lens is used for the correction of myopic eye as it can form real as well as virtual images of the objects placed in front of it. (Set- 31/1/1)

Q.46. In the following diagram showing dispersion of white light by a glass prism, the colours 'P' and 'Q' respectively are - (Set- 31/6/1)



(a) Red and Violet (b) Violet and Red

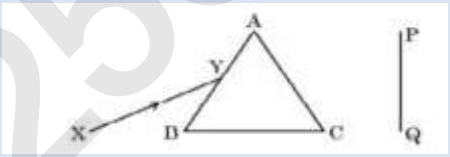
(c) Red and Blue (d) Orange and Green

Q.47. In human eye the part which allows light to enter into the eye is - (Set- 31/6/1)

(a) Retina (b) Pupil (c) Eye lens (d) Cornea

Section-B

Q.48. (a) A narrow beam XY of white light is passing through a glass prism ABC as shown in the diagram: Trace it on your answer sheet and show the path of the emergent beam as observed on the screen PQ. Name the phenomenon observed and state its cause. (Set- 31/1/1)



OR

(b) It is observed that the power of an eye to see nearby objects as well as far off objects diminishes with age.

(i) Give reason for the above statement.

(ii) Name the defect that is likely to arise in the eyes in such a condition.

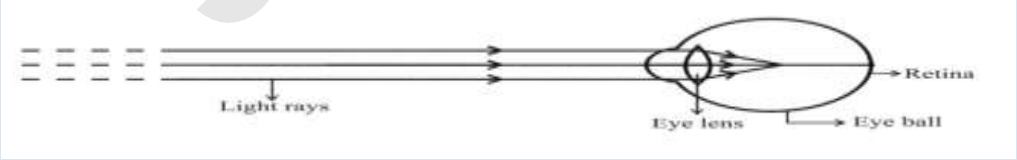
(iii) Draw a labelled ray diagram to show the type of corrective lens used for restoring the vision of such an eye.

Q.49. (a) When and where does a rainbow appear in the sky? Draw a labelled ray diagram to show its formation. (Set- 31/2/1)

OR

(b) What is scattering of light? Why does the clear sky appear blue?

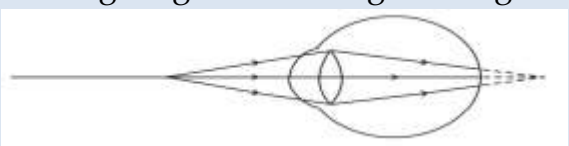
Q.50. Observe the following diagram and answer the questions following it: (Set- 31/4/1)



(i) Identify the defect of vision shown. (ii) List its two causes.

(iii) Name the type of lens used for the correction of this defect.

Q.51. Observe the following diagram showing an image formation in an eye: (Set- 31/5/1)



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(a) Identify the defect of vision shown in the figure.
(b) List its two causes and suggest a suitable corrective lens to overcome this defect.

SECTION C

Q.52. Give reasons for the following: (Set- 31/1/1)
(a) Danger signals installed at airports and at the top of tall buildings are of red colour.
(b) The sky appears dark to the passengers flying at very high altitudes.
(c) The path of a beam of light passing through a colloidal solution is visible.

Q.53. State one important function of the following parts of the human eye: (Set- 31/5/1)
(i) Retina (ii) Pupil
(b) State the role of ciliary muscles in focussing objects at varying distances from the eye.

Q.54. State reasons for Myopia. With the help of ray diagrams, show the (Set- 31/6/1)
(a) image formation by a myopic eye, and
(b) correction of myopia using an appropriate lens.

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